PYTHON ABL CASE STUDIES

Program Output Report

Outputs for all 10 case studies

Consolidated Output Document

Case Study 1

```
===== SCHOLARSHIP RESULTS =====
Rank: 1
Name: Priya
Aggregate: 92.0 %
Subject Average: 93.0 %
Status: Scholarship Allocated
Scholarship Amount: Rs. 20000
Rank: 2
Name: Ravi
Aggregate: 88.0 %
Subject Average: 86.0 %
Status: Scholarship Allocated
Scholarship Amount: Rs. 15000
Rank: 3
Name: Arjun
Aggregate: 80.0 %
Subject Average: 75.0 %
Status: Waitlisted
Total Eligible Applicants: 3
Scholarship Seats Available: 2
Selected Applicants: 2
```

Case Study 2

```
Month 1: 85 units, energy ■170.00, bill ■208.50
Month 2: 120 units, energy ■260.00, bill ■323.00
Month 3: 190 units, energy ■470.00, bill ■543.50
Month 4: 310 units, energy ■1050.00, bill ■1152.50
MONTH-TO-MONTH CHECK
Month 2: unusual increase of 54.9%
Month 3: unusual increase of 68.3%
Month 4: unusual increase of 112.1%
```

Case Study 3

```
EMPLOYEE SALARY STATEMENT
Employee: Ravi
Basic pay: ■24000.00
HRA: ■4800.00
Travel allowance: ■2400.00
Overtime pay: ■1200.00
Incentive: ■1500.00
Gross salary: ■33900.00
Deduction: ■2390.00
Net salary: ■31510.00
```

Case Study 4

ITEMIZED INVOICE
Notebook: 5 x ■80 = ■400.00; tax ■18.00
School Bag: 2 x ■900 = ■1800.00; tax ■194.40
Pen Pack: 3 x ■120 = ■360.00; tax ■16.20
Subtotal: ■2560.00
Discount: ■256.00
Tax: ■228.60
FINAL AMOUNT: ■2532.60

Case Study 5

```
Classes attended: 42
Total classes: 60
Current attendance: 70.00%
Attend the next 12 consecutive classes to reach 75%.
Attendance after recovery: 75.00%
```

Case Study 6

```
Accepted deposit: ■2000
Accepted withdrawal: ■1500
Rejected withdrawal: ■4000 (limit or balance exceeded)
Rejected: DEPOSIT ■0 (amount must be positive)
Accepted withdrawal: ■1000
ATM SUMMARY
Deposits: ■2000
Withdrawals: ■2500
Rejected transactions: 2
Closing balance: ■4500
```

Case Study 7

```
INVENTORY RECONCILIATION
Product: Pens
Expected closing stock: 35
Recorded closing stock: 35
Status: Matched
Replenishment required: 0 units
Product: Notebooks
Expected closing stock: 30
Recorded closing stock: 25
Status: DISCREPANCY (-5 units)
Replenishment required: 0 units
Product: Folders
Expected closing stock: 25
Recorded closing stock: 28
Status: DISCREPANCY (+3 units)
Replenishment required: 0 units
```

Case Study 8

```
STUDENT RESULTS
Asha: total=246, average=82.00
Bharat: total=215, average=71.67
Charan: total=145, average=48.33
Divya: total=274, average=91.33
SUBJECT AVERAGES
Python: 74.00
Maths: 71.25
English: 74.75
Top performer: Divya (91.33 average)
```

Case Study 9

```
Password: abc
Result: Invalid Password
- Password must contain at least 8 characters
- Missing uppercase letter
- Missing digit
- Missing special character
Password: Password123!
Result: Invalid Password
- Contains forbidden text: password
Password: Abc111!xyz
Result: Invalid Password
- Same character repeated 3 times consecutively
Password: Strong@42
Result: Valid Password
Password: Admin@1234
Result: Invalid Password
- Contains forbidden text: admin, 1234
```

Case Study 10

```
CAMPUS EVENT BUDGET
Fixed costs: ■25000
Variable cost per participant: ■180
Expected participants: 100
Total expected cost: ■43000
Sponsorship: ■18000
Registration contributions: ■25000
Total funds: ■43000
Funding surplus: ■0
Break-even registrations with current sponsorship: 100
```